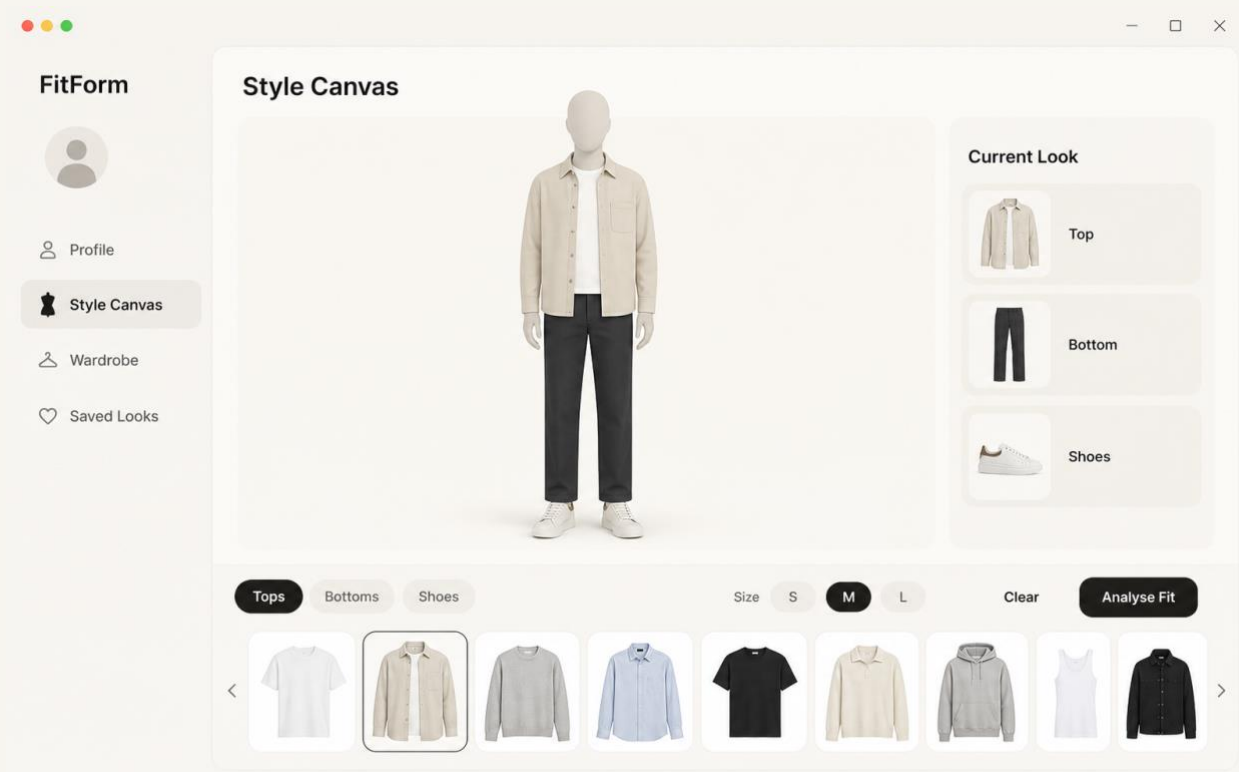


FitForm

Personalised 2D Outfit Visualiser

Course COMP 9003	Group 8	Members Avan Liu Lira Chen ZhanPing Zhou	Submission 30 August 2026
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1. Project idea

FitForm is a JavaFX desktop application that allows users to enter height, chest, waist and hip measurements, generate an approximate neutral 2D avatar and preview prepared clothing items on proportions closer to their own.

The system uses a hybrid method: it first classifies a broad body type using measurement ratios, then adjusts key avatar widths and lengths with the user's actual measurements. Clothing is selected from a curated transparent-PNG wardrobe and positioned using simple shoulder, waist and hip anchors.

Core innovation: unlike a standard wardrobe application, FitForm combines measurement-responsive 2D visualisation with a Java-based fit score and size recommendation, without claiming realistic 3D or cloth simulation.

2. Idea inspiration

The idea responds to a common online-shopping problem: garments are usually shown on a limited range of models, making it difficult for users to imagine the same outfit on different proportions. Commercial virtual try-on tools can require computer vision, AI or 3D assets, so our project explores a deliberately simpler and more explainable alternative.

INSPIRATION	DESIGN RESPONSE
Generic product-model imagery	Generate a neutral avatar whose main proportions respond to user measurements.
Warm, accessible wardrobe apps	Use an ivory/beige JavaFX interface with large headings, rounded filters and clear clothing cutouts.
Need for technical depth	Use OOP, inheritance, collections, procedural shapes, body classification and fit-scoring algorithms.
Risk of over-complexity	Exclude 3D bodies, cloth physics, AI try-on, photo measurement and automatic image processing.

Proposed user flow

Profile > Generate avatar > Select top / bottom / shoes > Choose size > Analyse fit > Save look

3. Progress so far

At Week 4, the project remains in concept validation and feasibility planning. The team has defined the problem, selected JavaFX as the implementation platform, established the MVP, mapped the main Java classes and developed a consistent four-screen UI direction.

AREA	CURRENT STATUS	NEXT VERIFIABLE OUTPUT
Concept	Defined: measurement-responsive neutral 2D outfit visualiser	Tutor validation of scope and project value
MVP	Profile, avatar, wardrobe, style canvas, fit result and saved looks	Acceptance criteria and prioritised backlog
Technical plan	JavaFX, procedural shapes, PNG overlays, OOP and collections	BodyProfile class and avatar proof of concept
UI direction	Four beige desktop screens designed for JavaFX implementation	FXML/Pane structure and shared CSS tokens
Testing	Success criteria drafted for classification, alignment and fit score	Sample profiles, expected outputs and user test script
Version control & hosting	GitHub repository initialized, branch protections configured, and GitHub Pages set up	Baseline project scaffold, shared Gradle/Maven build, and CI/CD workflow

Current challenges and responses

CHALLENGE	WHY IT MATTERS	PLANNED RESPONSE
Measurements cannot precisely reconstruct a person	People with similar measurements may have different shapes	Classify a broad body type, then adjust only key control points; label the result as approximate
Garment alignment can look inconsistent	Images may have different proportions or transparent margins	Use a curated asset library and fixed anchors for each garment category
Realistic fit is technically complex	Cloth drape and collision require advanced graphics	Evaluate visual proportion and size ranges, not physical cloth behaviour
Scope can expand quickly	Search, upload, weather and AI features could delay the MVP	Use preloaded items and click-to-select interactions; complete core acceptance tests first

4. Visual group work plan

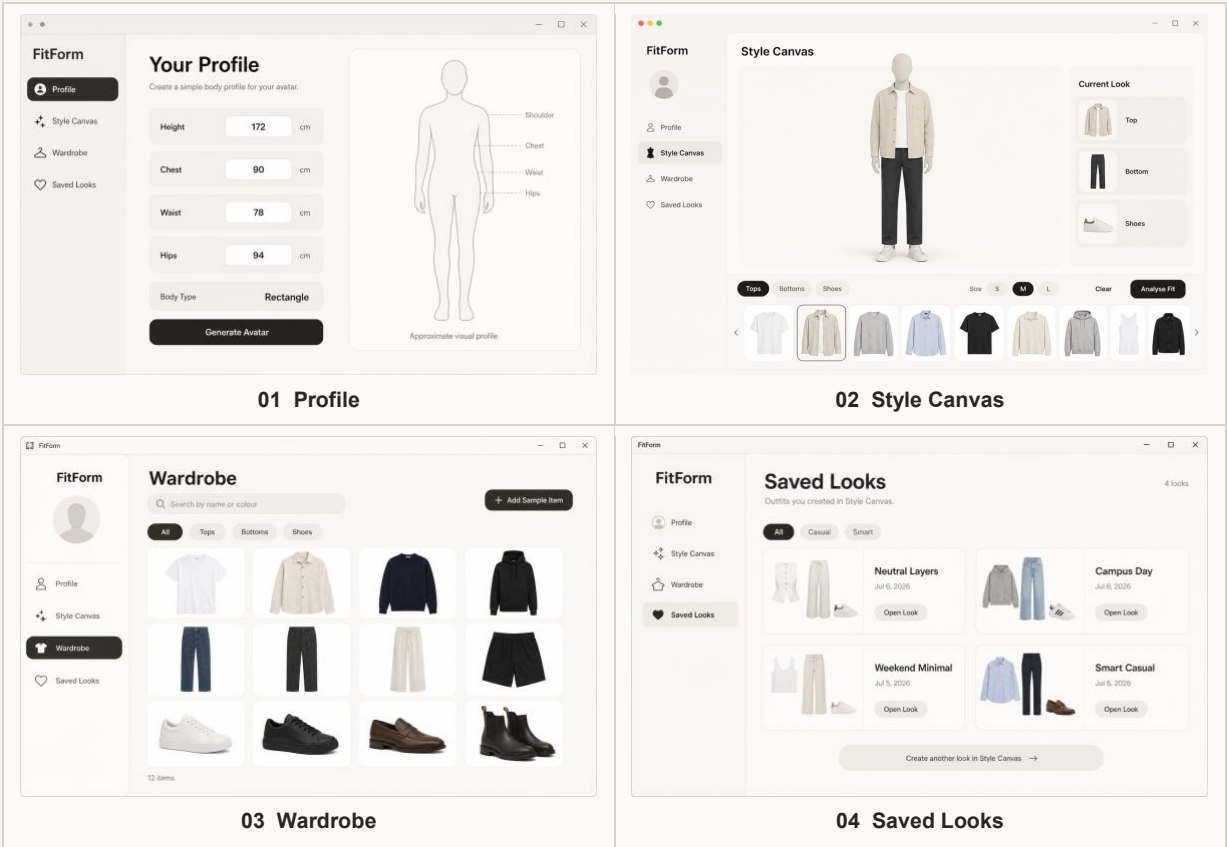
The project will use short build-test cycles. Each phase produces an observable output that can be demonstrated to the tutor, reducing the risk of spending too long on unvalidated features.

W4	W5	W6	W7	W8	W9+
Validate	Profile	Avatar	Wardrobe	Algorithms	Refine
Tutor feedback Scope + UI	BodyProfile Rules	JavaFX shape Response	PNG assets Anchors	Fit score Size	Testing Demo + report

Team responsibilities

WORKSTREAM	PRIMARY OWNER	SUPPORT	WEEKLY OUTPUT
Research + testing	Lira Chen/ Avan Liu	All members	User problem, test profiles and findings
UX + visual design	ZhanPing Zhou	All members	JavaFX screen layouts, CSS and assets
Java + algorithms	ZhanPing Zhou	All members	BodyProfile, classification and fit score
Avatar + wardrobe	Avan Liu / Lira Chen	All members	Procedural shape, anchors and PNG overlays
Coordination + report	All members	All members	Backlog, meeting notes and evidence

UI concept gallery



JavaFX implementation mapping

SCREEN	SIMPLE JAVAFX STRUCTURE
Profile	HBox + VBox input rows + Canvas/Pane avatar preview
Style Canvas	BorderPane + ImageView layers + horizontal ScrollPane
Wardrobe	TilePane/GridPane inside ScrollPane using preloaded assets
Saved Looks	ArrayList<Outfit> rendered as repeated HBox/VBox items